

(1.1)

V nad $\mathbb{C}$ skalární součin $\langle, \rangle$

$$\Rightarrow \forall \vec{u}, \vec{v} \in V \quad 1) \operatorname{Re}(\langle \vec{u}, \vec{v} \rangle) = \frac{1}{4} (\|\vec{u} + \vec{v}\|^2 - \|\vec{u} - \vec{v}\|^2)$$

$$2) \operatorname{Im}(\langle \vec{u}, \vec{v} \rangle) = \frac{1}{4} (\|\vec{u} - i\vec{v}\|^2 - \|\vec{u} + i\vec{v}\|^2)$$

D: 1) $\|\vec{u} + \vec{v}\|^2 - \|\vec{u} - \vec{v}\|^2 = \langle \vec{u} + \vec{v}, \vec{u} + \vec{v} \rangle - \langle \vec{u} - \vec{v}, \vec{u} - \vec{v} \rangle =$ podle vlastnosti skal. součinu =

$$= \langle \vec{u}, \vec{u} \rangle + \langle \vec{u}, \vec{v} \rangle + \langle \vec{v}, \vec{u} \rangle + \langle \vec{v}, \vec{v} \rangle - \langle \vec{u}, \vec{u} \rangle + \langle \vec{u}, \vec{v} \rangle + \langle \vec{v}, \vec{u} \rangle - \langle \vec{v}, \vec{v} \rangle$$

$$= 2\langle \vec{u}, \vec{v} \rangle + 2\langle \vec{v}, \vec{u} \rangle = 2 \left(\underbrace{\langle \vec{u}, \vec{v} \rangle + \overline{\langle \vec{u}, \vec{v} \rangle}}_{2\operatorname{Re}(\langle \vec{u}, \vec{v} \rangle)} \right) = 4 \operatorname{Re}(\langle \vec{u}, \vec{v} \rangle)$$

a tedy $\operatorname{Re}(\langle \vec{u}, \vec{v} \rangle) = \frac{1}{4} (\|\vec{u} + \vec{v}\|^2 - \|\vec{u} - \vec{v}\|^2)$

2) $\|\vec{u} - i\vec{v}\|^2 - \|\vec{u} + i\vec{v}\|^2 =$ podobně = $\langle \vec{u} - i\vec{v}, \vec{u} - i\vec{v} \rangle - \langle \vec{u} + i\vec{v}, \vec{u} + i\vec{v} \rangle$

$$= \langle \vec{u}, \vec{u} \rangle - i\langle \vec{u}, \vec{v} \rangle - \overline{i}\langle \vec{v}, \vec{u} \rangle - \langle \vec{v}, \vec{v} \rangle - \langle \vec{u}, \vec{u} \rangle - i\langle \vec{u}, \vec{v} \rangle - \overline{i}\langle \vec{v}, \vec{u} \rangle + \langle \vec{v}, \vec{v} \rangle =$$

$$= -2i \left(\langle \vec{u}, \vec{v} \rangle + \langle \vec{v}, \vec{u} \rangle \right) = -2i \left(\underbrace{\langle \vec{u}, \vec{v} \rangle + \overline{\langle \vec{u}, \vec{v} \rangle}}_{2i\operatorname{Im}(\langle \vec{u}, \vec{v} \rangle)} \right) = -4i^2 \operatorname{Im}(\langle \vec{u}, \vec{v} \rangle) = 4 \operatorname{Im}(\langle \vec{u}, \vec{v} \rangle)$$

a tedy $\operatorname{Im}(\langle \vec{u}, \vec{v} \rangle) = \frac{1}{4} (\|\vec{u} - i\vec{v}\|^2 - \|\vec{u} + i\vec{v}\|^2)$ □